MOHAMED SAAD

Digital Solutions Developer | Full Stack & Backend Engineer

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PROFESSIONAL SUMMARY

Results-driven **Digital Solutions Developer** with deep expertise in architecting, developing, and deploying scalable web applications and enterprise-grade REST APIs. Specialized in **Java, Spring Boot, Spring Security, React.js, and Next.js** with extensive experience designing resilient backend systems, secure JWT authentication mechanisms, normalized PostgreSQL databases, and integrating modern **AI models (Google Gemini API & vector embeddings)** to resolve business-critical bottlenecks.

TECHNICAL SKILLS

Languages: Java (8/17/21), TypeScript, JavaScript, SQL, HTML5, CSS3

Backend & APIs: Spring Boot, Spring Security, Spring WebFlux, Hibernate/JPA, RESTful APIs, JWT Auth, Microservices

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Responsive Web Design, State Management

Databases: PostgreSQL, pgvector (Vector Embeddings), MySQL, Redis, Supabase, Query Optimization

DevOps & Cloud: Docker, Docker Compose, Git & GitHub Actions (CI/CD), Maven, npm, Nginx, Linux CLI

AI & Automation: Google Gemini AI API, Semantic Search, Automated CV/Resume Parsing, Telegram Bot API

WORK EXPERIENCE

Jan 2024 – Present

Full Stack Software Engineer • JobFinder Platform

- Architected and developed a full-stack job search and matching platform using **Spring Boot** and **Next.js / React**.
- Implemented secure session and token handling utilizing **JWT authentication**, Spring Security, and HttpOnly cookies.
- Resolved Tomcat thread exhaustion and database connection starvation by refactoring long-running AI extraction routines into reactive asynchronous streams using **Spring WebFlux**.
- Integrated **Google Gemini AI** for automated CV parsing, smart skill extraction, and semantic candidate-to-job matching using **pgvector** vector embeddings in PostgreSQL.
- Designed normalized relational database schemas with indexed queries, reducing candidate search response times under 150ms.

Jun 2022 – Dec 2023

Backend Developer • Tech Solutions Ltd.

- Developed and maintained high-throughput RESTful APIs for commercial web applications using **Spring Boot** and **PostgreSQL**.
- Diagnosed and eliminated Hibernate N+1 query bottlenecks and optimized database transactions, enhancing overall API throughput by **40%**.
- Containerized application microservices and supporting dependencies using **Docker & Docker Compose** for seamless staging and deployment.
- Established automated code quality checks and participated in peer code reviews upholding clean code and SOLID architectural principles.

KEY ENGINEERING PROJECTS

JobFinder Platform — Java, Spring Boot, Spring WebFlux, PostgreSQL (pgvector), Next.js, Docker

- Engineered an intelligent career platform with real-time job application tracking, CV skill parsing, and vectorized search.

Expense Tracker Bot — Java, Spring Boot, Telegram Bot API, PostgreSQL

- Created an automated financial management Telegram bot providing instant command routing, categorization, and weekly analytics.

Ahl Masr Restaurant Digital Platform — React, TypeScript, Tailwind CSS, Framer Motion

- Developed a high-converting digital ordering menu and restaurant showcase with real-time category filtering and WhatsApp API integration.

EDUCATION & CORE PRINCIPLES

Bachelor's Degree in Computer Science / Information Systems • Cairo, Egypt
Core Principles: Clean Architecture, Test-Driven Quality, High Availability, Scalable Database Design, Secure Coding.